

The Role of Urban Greenspace in Shaping Labor Market Outcomes

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Job Market Paper

January 10, 2026

Abstract

This paper examines how urban greenspace influences wages and local economic dynamics within a spatial-equilibrium framework. Using panel data for 950 U.S. urban counties from 2011 to 2019 and an instrumental-variables strategy based on historical land cover, I show that greener cities exhibit significantly lower wages, consistent with compensating differentials. At the same time, greener cities experience faster population and employment growth over a three-year horizon, indicating that greenspace attracts households and expands labor supply. These results demonstrate that greenspace is not only an environmental amenity but also an economic force shaping wage levels, migration patterns, and regional development. This suggests that urban greenspace can be a viable lever for local economic policy, influencing where people live and work.

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1 Introduction

Urban greenspace is often viewed as a desirable feature of sustainable cities, credited with improving public health (WHO, 2016), mitigating heat (Gunawardena et al., 2017), managing stormwater (Berland et al., 2017), and enhancing overall livability (Reyes-Riveros et al., 2021). However, in cities where land is scarce and development pressures are high, policy-makers and planners frequently face trade-offs between allocating land for greenspace versus using it for higher-yielding commercial or residential purposes. This dilemma echoes broader public investment trade-offs, faced by both advanced and developing economies, where infrastructure needs often crowd out environmental amenities (Gurara et al., 2017). In such contexts, greenspace is typically treated as a secondary amenity, valued in principle but de-prioritized in practice. This study offers a concrete economic rationale for investment in urban greenspace. By showing that workers are willing to accept lower wages in exchange for access to greenspace, it quantifies a compensating differential that can help justify urban greening efforts not merely on environmental or aesthetic grounds, but as a strategy for enhancing a city’s economic competitiveness and appeal. At the same time, greenspace affects not only wages but also population and employment patterns. If greenspace raises residential utility, households should gradually relocate toward greener areas, increasing population and, over time, expanding local employment. This study therefore complements the wage-based compensating differential with evidence on population and employment dynamics associated with greenspace.

This study asks a simple but underexplored question: *do urban workers accept lower wages in exchange for living in greener areas?* In other words, is greenspace a local amenity that increases residential utility sufficiently for workers to trade off earnings in return for improved environmental quality? According to the compensating differentials framework (Roback, 1982; Rosen, 1979), local wages reflect not only productivity but also the value of amenities that enhance utility. If greenspace raises utility, then, in equilibrium, wages should be lower in greener counties as workers are willing to accept reduced earnings to

locate in higher-amenity environments. While such non-market valuation frameworks have been applied to value amenities like climate (Albouy et al., 2016), air quality (Chay and Greenstone, 2003; Currie et al., 2015), and crime (Gyourko and Tracy, 1991; Dentler and Rossi, 2024), analogous evidence for urban greenspace remains limited. Understanding this wage–amenity relationship provides the first step toward quantifying the economic value of greenspace within a revealed-preference framework.

A related question is whether this amenity advantage leads to demographic and employment adjustments. A large literature on natural amenities and quality of life (QOL) shows that households systematically migrate toward places with higher amenity levels, such as favorable climate and landscape features. This behavior helps explain persistent patterns of population growth across U.S. regions (Chen and Rosenthal, 2008; Rickman and Wang, 2017). Similar dynamics are observed internationally, where environmental amenities or stressors influence migration patterns (Beltran and Hadzi-Vaskov, 2023). Reviews of the amenity-driven growth literature similarly conclude that landscape and environmental amenities can promote local population growth and, in some cases, stimulate income and employment, even though the strength of these effects varies by context (Waltert and Schläpfer, 2010; Weinstein et al., 2023). Building on this work, I test whether greener urban counties experience faster increases in population and employment than their less green counterparts, consistent with a spatial-equilibrium mechanism in which workers relocate toward high-amenity locations and firms respond to the resulting expansion in local labor supply. These findings are relevant for urban planners and policymakers considering investments in public amenities, emphasizing the role of regional development strategies and spatial disparities in shaping long-term growth (Floerkemeier et al., 2021). The results suggest greenspace isn’t just preferred for aesthetics or health, but can influence labor market equilibrium and growth.

To examine the utility value of greenspace, I use a panel dataset covering 950 urban counties in the United States (U.S.) from 2011 to 2019. My measure of greenspace is the

Normalized Difference Vegetation Index (NDVI), a remote-sensing index derived from satellite imagery that captures the extent and density of green vegetation, widely used in other settings (Huang et al., 2021; Zheng et al., 2021; De la Iglesia Martinez and Labib, 2023; Davis et al., 2023) . NDVI offers high spatial and temporal resolution, allowing for consistent, comparable tracking of greenspace across counties over time. I combine this with county-level wage data from the U.S. Bureau of Labor Statistics (BLS) and a rich set of controls, including measures of rents, industrial composition, and socio-economic characteristics, to estimate the relationship between greenspace and wages, population and employment. To address potential endogeneity concerns, I employ an instrumental variable (IV) strategy using historical land cover data from the 1970s-80s, leveraging the persistent nature of land use patterns to isolate exogenous variation in contemporary greenspace.

The results indicate a robust and statistically significant negative relationship between greenspace and urban wages, consistent with the hypothesis that workers value greener environments and are willing to trade off earnings to access them. Beyond wages, I also examine demographic and employment adjustments by estimating three-year changes in population and employment. The level-difference specifications show that greener counties experience significantly larger increases in population and employment over a three-year horizon. These effects are consistent with the spatial-equilibrium framework in which wages are lower in more attractive places. But persistent movement toward equilibrium also means that higher amenity places have population growth, resulting in employment growth as firms expand or relocate in response to higher labor supply (Roback, 1982; Glaeser and Gottlieb, 2009).

By documenting the wage–amenity tradeoff and the associated demographic and employment adjustments, this paper makes several contributions to the literature. First, it extends the compensating differentials framework to a widely recognized but understudied amenity such as greenspace. Much of the existing work on greenspace relies on perceived measures obtained from qualitative surveys or self-reported well-being assessments (Laforteza et al.,

2009; Kabisch, 2015). While these are valuable, they do not necessarily reflect actual tradeoffs in labor markets. In contrast, I adopt a revealed-preference approach in which wages, population flows, and employment levels all respond to amenity differences in a manner consistent with spatial equilibrium theory (Roback, 1982; Glaeser and Gottlieb, 2009). Second, while prior environmental economics research has predominantly focused on the capitalization of greenspace into housing prices (Cho et al., 2008; Geoghegan, 2002; Zambrano-Monserrate et al., 2021; Zhang et al., 2020), my wage-based estimates provide an independent and theoretically coherent valuation of greenspace that is not mediated through housing markets. Third, by leveraging historical land cover as an instrument for contemporary greenspace, I provide one of the first causal estimates linking urban vegetation to labor-market outcomes and demographic adjustments. This IV strategy addresses the endogeneity concerns, such as reverse causality, that commonly limit inference in greenspace valuation studies. Finally, unlike many natural amenities such as coastline, mountains, or climate, greenspace is policy malleable. Tree planting, park development, and land-use regulations can substantially increase greenspace within relatively short time horizons. As such, my findings highlight the potential for urban greening policies to enhance livability, support population retention, attract labor, and stimulate employment growth.

The remainder of this paper is organized as follows. The *Previous Literature* section reviews the existing work related to natural amenities and their influence on regional labor markets. The *Theoretical Framework* section outlines the mechanism through which greenspace affects wages, and in turn, population flows and employment. The *Data* section describes the construction of the greenspace measures and the accompanying socioeconomic and geographic variables. The *Methodology* section presents the empirical specifications and explains the identification strategy. The *Results* section reports the main estimates, and the *Discussion and Policy Implications* section interprets these findings in the context of urban development and regional growth. The paper ends with a *Conclusion* section summarizing the key contributions and implications for future research.

2 Previous Literature

The spatial-equilibrium and compensating-differentials framework developed by Roback (1982) and Rosen (1979) provides the standard lens for valuing local attributes. In this framework, wages and housing costs adjust so that, in equilibrium, utility is equalized across space. This approach underpins a large literature that infers amenity values from observed wage-rent patterns. Additionally, a substantial body of research has focused on how environmental amenities influence housing prices, land values, and related real-estate outcomes (Anenberg and Kung, 2020; Harris, 2024; Glaeser et al., 2014; Bischoff, 2012; Kahn and Tracy, 2024). The framework predicts that both wages and housing values adjust to reflect the attractiveness of a place. The degree to which these adjustments appear in the labor or housing markets depends on how amenities affect household utility, firm productivity, or both (Roback, 1982, 1988).

A complementary literature, tracing back to the 1970s, (Polinsky and Shavell, 1976; Yinger, 1976) conceptualizes cities as bundles of consumption opportunities. In this view, households choose locations partly to access amenity sets that are difficult to replicate privately. Glaeser et al. (2001) formalize the idea of the “consumer city,” and Glaeser and Gottlieb (2006) emphasize how advances in transportation and communication loosened the primacy of proximity-to-jobs, increasing the salience of consumption amenities such as cultural institutions, varied dining, entertainment, school quality, and neighborhood safety. This body of work uses inter-city comparisons to show that amenity differences help explain where people choose to live and, through equilibrium adjustments, why wages and rents differ across places. In short, amenities are not peripheral, rather, they are central to urban sorting and to the equilibrium prices that studies use to infer quality-of-life (QOL) differences (Glaeser et al., 2001; Glaeser and Gottlieb, 2006).

Environmental attributes are a prominent component of the amenity bundle examined in QOL studies. Climatic factors, water access, and resource endowments have been analyzed within compensating-differentials frameworks to explain systematic regional differences in

outcomes (Albouy et al., 2016). Recent work highlights how natural amenities and resource settings can shape development trajectories and location choices, reinforcing the role of environment in spatial equilibrium (Clay et al., 2023). At the urban scale, the presence of natural amenities is linked to growth by improving a city’s ability to attract households and firms (Stephens and Partridge, 2015; Hausman et al., 2025; Weinstein et al., 2023), thus, connecting directly back to the QOL literature. When environmental conditions raise residential utility or affect production, equilibrium wages and rents adjust in ways that reveal the value of those conditions (Roback, 1982; Rosen, 1979).

Within environmental amenities, urban greenspace such as parks, tree-lined streets, gardens, and other associated features, have been studied extensively for their contributions to physical and mental health, air-quality exposure, and social well-being (Carrus et al., 2015; Wolf et al., 2020; Donovan and Gatzliolis, 2019; Weichenthal et al., 2016; Gu et al., 2022). Greenspace can also function as “third places,” facilitating interaction and community ties (Kabisch, 2015), and they provide non-market benefits that are salient to household location choices (Lafortezza et al., 2009). Empirically, much of this literature documents capitalization of greenspace into housing prices and land values (Siriwardena et al., 2025; Pandit et al., 2014), and evaluates broader cost–benefit implications of urban trees and related investments (Soares et al., 2011). Under the spatial-equilibrium lens, greenspace is treated as a consumption amenity, shifting residential utility rather than firm productivity. Unlike production-linked attributes such as access to waterways, greenspace is also primarily shaped by urban policy. Under that interpretation, compensating differentials may manifest in both wages and rents consistent with Roback (1982, 1988) and more recent literature (Chen and Wang, 2013).

Taken together, the QOL, amenities, and environmental-economics literature establish that local attributes, especially environmental ones, have measurable value revealed in equilibrium prices and quantities (Roback, 1982; Rosen, 1979; Albouy et al., 2016; Weinstein et al., 2023). However, within the greenspace literature specifically, the emphasis has largely

been on health and social outcomes and on housing-market capitalization (Carrus et al., 2015; Wolf et al., 2020; Donovan and Gatzolis, 2019; Kabisch, 2015; Siriwardena et al., 2025; Pandit et al., 2014; Soares et al., 2011). By contrast, direct evidence on the contribution of urban greenspace to wage differentials, or broader labor market outcomes, remains comparatively limited. Clarifying that wage channel within the established spatial-equilibrium framework is the natural next step implied by this literature.

A parallel literature investigates how natural and urban amenities shape population change, emphasizing that households systematically migrate toward high-amenity locations. Amenities such as climate, landscape features, environmental quality, and recreational access strongly influence long-run population growth patterns across regions (Chen and Rosenthal, 2008). Empirical work shows that, even after controlling for wages and rents, amenity-rich counties attract more in-migration and retain population more effectively, reflecting households' preferences for non-market attributes (Weinstein et al., 2023; Knapp and Graves, 1989). While in equilibrium, utility is equalized across space, a large literature suggests that there may be persistent movement toward equilibrium reflected in population changes. In the context of natural amenities, studies demonstrate that environmental attributes help explain divergent regional trajectories in population growth and household sorting across the U.S. (Rickman and Wang, 2017). Despite the depth of this literature, the role of urban greenspace, as opposed to broader natural amenities, has received comparatively little attention, especially in terms of its revealed-preference effects on population change.

A related body of research examines how amenities shape employment dynamics, particularly through labor supply shifts. Amenities influence not only where households choose to live but also where firms choose to operate, especially when amenities enhance worker recruitment, retention, and productivity (Glaeser and Gottlieb, 2009). Empirical evidence shows that natural amenities and environmental quality can stimulate employment growth by attracting skilled labor, supporting consumer-service industries, and encouraging firm formation in high-amenity regions (Partridge and Rickman, 2008). Within the spatial-equilibrium

framework, as labor supply increases, firms adjust location decisions, hiring, and capital deployment. Research specifically linking urban greenspace to employment growth remains limited, leaving open the question of whether greener urban areas experience systematically stronger employment expansion as part of the broader equilibrium adjustment to environmental amenities.

3 Theoretical Foundation

The analysis builds on the spatial-equilibrium framework of Roback (1982), in which workers choose locations by comparing wages, housing costs, and local amenities. If a local amenity affects workers, not firms, then we can examine the value of the amenity through its impact on wages.

A simplified indirect utility function for a representative worker in region i is:

$$V_i = V(w_i, r_i, s_i),$$

where w_i is the local wage, r_i is the rental cost of housing, and s_i represents local amenities, including urban greenspace. I assume standard monotonicity conditions in the indirect utility function: utility increases in wages and local amenities, and decreases in housing costs.

$$\frac{\partial V}{\partial w_i} > 0, \quad \frac{\partial V}{\partial r_i} < 0, \quad \frac{\partial V}{\partial s_i} > 0.$$

Under high mobility, workers relocate until:

$$V_i = V_j \quad \forall i, j,$$

meaning wages and rents adjust to equalize utility across space. Because greenspace raises utility, spatial equilibrium requires either lower wages or higher rents in greener regions. The

resulting compensating differential satisfies:

$$\frac{\partial w_i}{\partial s_i} = -\frac{\partial V/\partial s_i}{\partial V/\partial w_i},$$

implying that wages decline as amenities improve, holding utility constant.

At the same time, while lower wages are the equilibrium outcome of this model, Rappaport (2004) and others have shown that spatial equilibrium is the long-run steady state of a growth model, and small frictions can make migration toward equilibrium persistent. Thus, following Stephens and Partridge (2015), if migration is ongoing, then one can expect to see higher levels of population growth in places with more greenspace. This is in line with Rappaport (2007), Partridge et al. (2012) and others, showing that population change provides a good proxy for migration. Additionally, changes in employment can also provide some evidence of the value of greenspace as an amenity in moving toward equilibrium. Household movements affect labor supply and firm movements affect labor demand, jointly determining a region’s employment level. Therefore, even if there is no change in labor demand, an increase in labor supply will increase employment levels.

Guided by this framework, I test whether greener counties exhibit (i) lower equilibrium wages and (ii) higher subsequent growth in population and employment, patterns consistent with workers valuing greenspace and sorting toward high-amenity locations.

4 Data

My analysis uses data on 950 urban counties in the contiguous 48 U.S. states (excluding Alaska and Hawaii) for the years 2011 - 2019. For classification of urban and rural, I rely on the identification provided by the Office of Management and Budget (OMB, 2023).

The academic literature has defined greenspace in various ways. A systematic review of the broad academic literature across multiple disciplines was conducted by Taylor and Hochuli (2017) and finds two major definitions of greenspace. For purposes of this study,

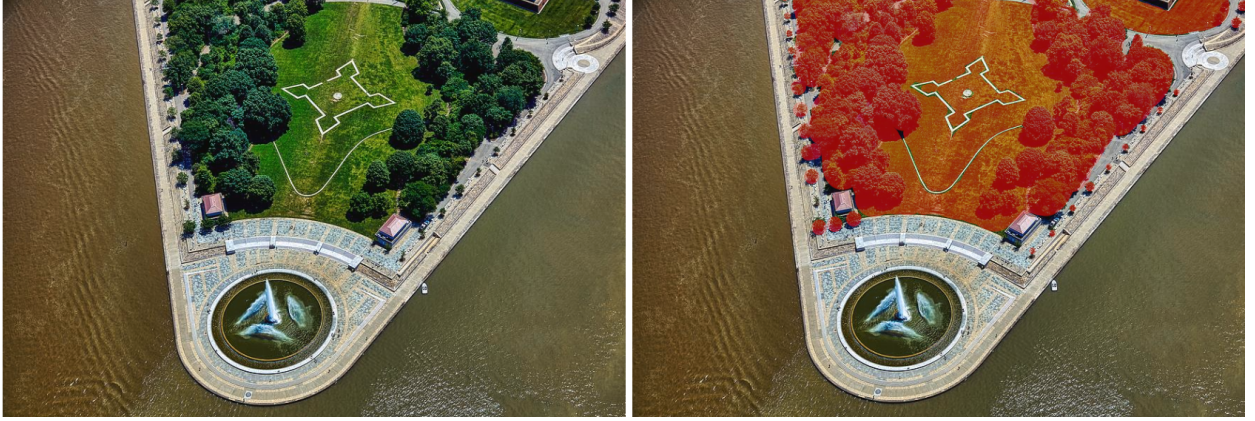


Figure 1: NDVI Vision

I focus specifically on urban vegetation, both public and private, encompassing features such as trees, parks, gardens and yards typically referring to vegetated forms of open space within urban environments (Chong et al., 2013; Bastian et al., 2012). This is also consistent with the definition used by the U.S. Environmental Protection Agency (EPA), which defines greenspace as urban land that is partly or completely covered with grass, trees, shrubs, or other vegetation (EPA, 2023).

I measure greenspace using the MODIS NDVI product (MCD13Q1 v061), which provides 16-day composites at 250-meter resolution from the Terra and Aqua satellites. For each year 2011–2019, I mosaic the June–July scenes over the contiguous U.S., apply the product’s quality flags to mask clouds/snow, and reproject to NAD83 / CONUS Albers (EPSG:5070). I then aggregate NDVI to my analysis unit of counties and construct a consistent, area-appropriate panel of summer vegetation greenness. *Figure 1* shows a visual representation of what NDVI identifies as greenspace, highlighted in red, from the satellite image of Point State Park in Pittsburgh, Pennsylvania.

Although there are many vegetation indexes, the NDVI is preferred by many disciplines as it enables comparison across spatial and temporal spectrum (Huang et al., 2021) and thus has been widely used in analyzing greenspace in other disciplines (Donovan and Gatzolis, 2019; Huang et al., 2021; Lei et al., 2021; Davis et al., 2023; Zheng et al., 2021). The NDVI uses the Near Infra-Red band and the red band spectrums of satellite images to

calculate an index value with a range of -1 to +1, with the higher the value, the higher the greenspace, values below 0 are associated with non-greenspace (*equation 1* shows the calculation formula).

$$\text{NDVI} = \frac{\text{NIR} - R}{\text{NIR} + R} \quad (1)$$

The satellite images were read into a GIS software, and were superimposed on a county shapefile. Using a raster extractor, the county level NDVI values were extracted as a mean of the all the NDVI indexes value grids within a county shape polygon. The previous literature also shows that NDVI values can fluctuate seasonally, however, they are highest during the summer (Jia et al., 2004; Kulenbekov et al., 2021). At the same time, satellite images can often be compromised by cloud covers, making a one-off assessment faulty. As a result, to create annual measures of greenspace, I used the bi-weekly satellite images for the months of June and July for each year while also correcting for cloud or snow covers. In addition, My analysis exclusively considered values exceeding 0 when extracting data from the raster files. My sample analysis reveals that the mean daily average NDVI value is 0.343 per county per annum, while the mean daily maximum value is 0.848 per county per annum, as depicted in *table 1*.

As described below, the primary dependent variable in the analysis is county-level wages, obtained from the Quarterly Census of Employment and Wages (QCEW) from the BLS. To examine broader labor market dynamics, I also analyze two additional outcomes: changes in total population level over 3 years, sourced from the U.S. Census Bureau, to evaluate whether greenspace attracts more residents. Changes in 3-year employment levels, collected from BLS, were also used to assess whether greener areas exhibit higher employment growth. *Table 1* presents descriptive statistics for all variables included in the model. On average, counties report annual wages of approximately \$42,000.

Table 1: Descriptive statistics of key variables.

Variable	N	Mean	SD	Min	Max
Average Annual Wage	10,750	41,721	10,339	20,968	138,273
Annual average of daily mean NDVI	10,750	0.343	0.142	0.00481	0.848
Annual average precipitation	10,750	0.111	0.0387	0.00250	0.282
Annual High Temperature days	10,750	30.66	16.97	0	105
% of people employed	10,750	57.41	6.92	21.40	77.60
% of population with Highschool Degree	10,750	62.16	8.038	19	81.20
% of population with Bachelors Degree	10,750	15.84	5.955	2.600	38
% of population with Graduate Degree	10,750	8.986	4.755	0.400	40.30
% of population less than HS	10,750	13.01	5.812	1.300	49.50
Total Population	10,750	238,739	516,492	709	1.011e+07
% of people White	10,750	82.14	14.41	12.60	99.80
% of people Black	10,750	9.949	12.59	0	78.50
% of people American Indian/Alaskan	10,750	0.798	3.191	0	83.90
% of people Asian	10,750	2.161	3.263	0	36.50
% of people Hawaii/Pacific	10,750	0.0752	0.179	0	2.300
% of people other race	10,750	2.495	3.624	0	38.40
% of people multiracial	10,750	2.374	1.399	0	18.80
% of people working in-county	10,750	59.64	19.94	11	98.50
Miles to nearest MSA	10,750	15.79	12.70	0	62.15
Median Rent	10,750	916.46	266.15	497	3269
House Price Index (HPI-86 base prices)	10,539	363.41	240.56	79.36	2575
County land area (sq km)	10,750	2.194e+09	3.438e+09	5.868e+07	5.195e+10
NAICS_11 Agriculture	10,140	2.721956	4.805812	0	68.58669
NAICS_21 Mine, Oil, Gas	10,140	0.6705086	2.413327	0	57.22946
NAICS_22 Utilities	10,140	0.5519818	1.224466	0	24.88364
NAICS_23 Construction	10,140	6.324081	2.794917	0.2606158	44.29266
NAICS_31 Manufacturing 1	10,140	1.879054	3.225592	0	40.32463

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Variable	N	Mean	SD	Min	Max
NAICS_32 Manufacturing 2	10,140	2.979599	3.085709	0	30.57712
NAICS_33 Manufacturing 2	10,140	5.257561	5.236238	0	64.09611
NAICS_42 Wholesale	10,140	3.054581	1.648293	0	20.899
NAICS_44 Retail 1	10,140	5.094034	1.472625	0	13.71592
NAICS_45 Retail 2	10,140	5.970427	1.998292	0.0308603	37.42886
NAICS_48 Transport 1	10,140	2.527547	1.957322	0.046642	33.96606
NAICS_49 Transport 2	10,140	0.9493777	2.055554	0	55.95381
NAICS_51 Information	10,140	1.094488	0.8930219	0	9.724973
NAICS_52 Finance	10,140	2.993985	2.229393	0.0004737	41.56274
NAICS_53 Real Estate	10,140	1.216823	0.6523989	0	8.1391
NAICS_54 Professional, Science Tech	10,140	3.952043	3.461146	0.112946	81.84882
NAICS_55 Management	10,140	0.8352158	1.292226	0	29.70073
NAICS_56 Admin and Waste	10,140	4.722905	2.19046	0.1653116	21.49744
NAICS_61 Education	10,140	1.736387	2.136355	0	35.575
NAICS_62 Healthcare, Social Assistance	10,140	11.50715	4.487855	0.1153433	50.20877
NAICS_71 Arts, Entertainment, Food	10,140	1.564459	2.089524	0	47.60313
NAICS_72 Accommodation, Food service	10,140	8.063525	3.585911	0	70.47432
NAICS_81 Other Services	10,140	5.441253	1.48685	0.3166605	29.79453
NAICS_90 Government	10,140	18.837	8.280837	0.2606457	89.59314
NAICS_99 Others	10,140	0.0540546	0.1146866	0	1.167113
Hist_NAICS Agriculture	10,140	67,182.26	196,605.2	296	3,391,725
Hist_NAICS Mining	10,140	59,286.09	180,298.8	205	3,022,557
Hist_NAICS Utilities	10,140	7,896.164	17,658.47	70	369,168
Hist_NAICS Construction	10,140	975.4112	836.0193	0	8,467
Hist_NAICS Manufacturing	10,140	6,920.752	17,618.17	30	367,024
Hist_NAICS Wholesale	10,140	1,612.758	1,923.212	0	25,156
Hist_NAICS Retail	10,140	65,569.50	196,380.9	221	3,383,801
Hist_NAICS Transport	10,140	53,962.73	169,040.4	133	2,940,851
Hist_NAICS AgSupport	9,830	353.3032	889.9014	1	17,391

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Variable	N	Mean	SD	Min	Max
Hist_NAICS OilGas	9,420	373.9947	1,488.075	0	29,495
Hist_NAICS MineSupport	10,070	3,296.813	8,025.062	1	128,165
Hist_NAICS Electric	10,040	14,992.67	46,916.36	1	824,507
Hist_NAICS Buildings	9,740	3,856.352	12,901.65	5	189,524
Hist_NAICS FoodMfg	9,830	3,495.74	14,149.56	0	236,258
Hist_NAICS WoodPaper	10,060	10,280.05	28,400.24	5	521,782
Hist_NAICS WholesaleDur	10,020	4,932.3	19,958.21	0	435,497
Hist_NAICS RetailGen	9,870	13,248.09	43,975.67	3	765,538
Hist_NAICS TruckWare	10,140	11,606.77	29,674.04	69	466,706
Hist_NAICS AirTrans	10,140	2,092.667	6,181.246	2	72,673
Hist_NAICS Rail	10,140	2,344.918	7,677.724	2	156,803
Hist_NAICS Passenger	10,140	7,169.186	20,514.44	62	394,880

I incorporate a comprehensive set of county-level characteristics as control variables to account for potential confounding factors. I incorporate indicators such as educational attainment levels, percentage of people working in the county, and racial composition that was derived from the American Community Survey (ACS) 5-year estimates. To account for the economic structure of each county, I include variables representing the industrial composition based on shares of employment at the two-digit North American Industry Classification System (NAICS) level. I also control for historical industrial composition using the Standard Industrial Classification (SIC) system from the 1970s-80s in my IV specifications. The control variables also include the county's land area, proximity to the center of the Metropolitan Statistical Area (MSA), annual high temperature days, and annual precipitation data obtained from the National Oceanic and Atmospheric Administration (NOAA).

My analysis focuses exclusively on urban counties, effectively comparing conditions across urbanized areas. A key challenge in this context is that many U.S. urban counties encompass significant rural land, such as forests, farmland, or undeveloped areas, that may be detected by NDVI as greenspace. However, these areas are often not directly accessible or enjoyed

by the urban population, potentially biasing the measure of greenspace. To address this concern, I include present-day land cover as a control variable in the model. This accounts for the influence of non-urban vegetation and isolates the effect of greenspace that is more likely to be experienced within the urban environment.

Finally, to control for the cost of living in an area, which can also affect utility, I control for the median rent using the Fair Market Rents (FMR) collected from the U.S. Department of Housing and Urban Development (HUD). The FMR provides median rents across different sizes of housing, which I average for each county per year.

5 Methodology

5.1 Empirical Model

My empirical strategy follows directly from the spatial-equilibrium framework outlined above. I estimate a series of Two-Stage Least Squares (2SLS) regression models that relate wages, population dynamics, and employment levels to county-level urban greenspace. The baseline specification (*equation 2*) examines how equilibrium wages vary with greenspace, while the extended models (*equation 3 & 4*) analyze adjustments in population and employment. In all cases, greenspace is measured using the NDVI, which ranges from 0 to 1.

5.1.1 Wage Specification

I begin by estimating the compensating differential implied by spatial equilibrium theory. The baseline wage regression takes the following log-linear form:

$$\log(\text{wage}_{ct}) = \beta_0 + \beta_1 \log(\text{NDVI}_{ct}) + \boldsymbol{\gamma}' \text{Controls}_{ct} + \theta_c + \tau_t + \varepsilon_{ct} \quad (2)$$

The dependent variable is the logarithm of average annual wages in county c and year t .

The coefficient of interest, β_1 , captures whether greener counties exhibit systematically lower wages, consistent with workers accepting lower compensation in exchange for amenity value. The vector Controls_{ct} includes socio-economic indicators, demographic structure, geographic characteristics, and industry composition. State fixed effects θ_c absorb time-invariant state characteristics, and year fixed effects τ_t control for national macroeconomic trends.

5.1.2 Population Change Specification

To assess whether greenspace influences demographic adjustments, I estimate a model where the dependent variable is the three-year change in population:

$$\Delta_3 \text{Population}_{c,t} = \alpha_0 + \alpha_1 \log(\text{NDVI}_{ct}) + \boldsymbol{\delta}' \text{Controls}_{ct} + \theta_c + \tau_t + u_{ct} \quad (3)$$

where $\Delta_3 \text{Population}_{c,t} = \text{Population}_{c,t} - \text{Population}_{c,t-3}$. The three-year difference is used for two reasons. First, population changes through migration are subject to mobility frictions, information constraints, and adjustment costs. A multi-year change smooths short-run fluctuations and better captures relocation patterns consistent with spatial equilibrium models. Second, using a three-year interval reduces noise from annual measurement error and avoids treating short-term volatility as meaningful demographic change.

5.1.3 Employment Change Specification

Additionally I estimate a parallel specification for employment dynamics. Employment levels adjust through firm entry, expansion, or contraction, processes that evolve more slowly than wage adjustments. The employment model takes the form:

$$\Delta_3 \text{Employment}_{c,t} = \kappa_0 + \kappa_1 \log(\text{NDVI}_{ct}) + \boldsymbol{\phi}' \text{Controls}_{ct} + \theta_c + \tau_t + \eta_{ct} \quad (4)$$

where $\Delta_3 \text{Employment}_{c,t}$ denotes the three-year change in total county employment. As with population, the three-year interval is theoretically justified because employment re-

sponses depend on firm location choices, capital investment, and local labor supply shifts, adjustments that are unlikely to occur within a single year, thus, a multi-year window provides a cleaner measure of economic adjustment.

All models are estimated using fixed effects to control for unobserved heterogeneity across counties and common shocks over time. However, due to the 3 year lag being used in the population and employment specification, the number of years in those sample is reduced. Across specifications, the coefficient on NDVI provides a unified measure of how urban greenspace affects (i) equilibrium wages, (ii) population changes, and (iii) employment growth.

5.2 Identification

One of the primary concerns with identifying the impact of greenspace on wages is that places with higher wages may have more fiscal flexibility (resulting from higher local taxes) to invest in more greenspace resulting in possible reverse causality. Greenspace may also be correlated with unobserved factors resulting in endogeneity concerns. Furthermore, there can be simultaneous effects on wages and greenspace due to factors like government zoning policies, which might make my estimators biased. To address these concerns, I employ an IV approach utilizing historical land cover data and controlling for historical industrialization, enabling me to isolate the causal impact of greenspace on labor outcomes. Specifically, I leverage land cover data from the 1970s-1980s, sourced from the United States Geological Survey (USGS), which predates the study period by approximately three decades. This temporal distance is crucial for my identification strategy as it is unlikely to be correlated with current wages given the significant economic restructuring over the last 50 years.

$$\log(\widehat{\text{NDVI}}_{ct}) = \alpha_0 + \alpha_1 \text{HistLC}_{ct} + \alpha_2 \text{HistIC}_{ct} + \gamma_i \text{Controls}_{ct} + \theta_c + \tau_t + \varepsilon_{ct} \quad (5)$$

Equation (5) specifies the first stage of an IV approach, where the dependent variable is the predicted log of NDVI, denoted as $\log(\widehat{\text{NDVI}}_{ct})$, for county c in year t . This formulation

leverages historical variables to isolate exogenous variation in urban greenspace, inspired from previous work (Stephens and Partridge, 2011). The term HistLC_{ct} represents historical land cover characteristics, such as forest or shrub density, which shape current greenspace levels but are plausibly uncorrelated with contemporary labor market shocks. HistIC_{ct} captures the historical industry composition of the county, reflecting long-run economic patterns that may influence the development and preservation of greenspace. Controls_{ct} includes additional observable county-level controls such as demographic, climatic, or spatial characteristics similar to the primary specification. The state fixed effects θ_c and year fixed effect τ_t account for unobserved state-specific heterogeneity and common temporal shocks, respectively, while ε_{ct} is the error term. This first-stage regression generates exogenous variation in NDVI used in the second-stage wage, population and employment equations to identify the causal effect of greenspace on labor market outcomes.

The selection of this instrument is predicated on two key assumptions. First, regarding relevance, I see a strong correlation between historic land cover patterns and contemporary NDVI values as evidenced by the first stage F-statistics (*see appendix 2*). Concerning the exclusion restriction, land cover from three decades prior is unlikely to exert a direct influence on current wage levels, except through its impact on present-day greenspace as measured by NDVI, when controlling for historical industrial composition. To test this, wages were regressed on historic land cover and find no statistical significance. These results are shown in *appendix 1*.

6 Results

6.1 Primary Results

The results in *table 2* provide empirical support for the hypothesized relationship between urban greenspace and labor market outcomes. Detailed regression results are provided in *appendix 3*. The instrumental-variables estimates, My preferred specification, reveal clear

and economically meaningful relationships between urban greenspace and county-level labor-market outcomes. Across all IV specifications, Log Greenspace (NDVI) is statistically significant with the expected sign, supporting the interpretation that greenspace affects wages, population flows, and employment levels as predicted through the spatial-equilibrium mechanism.

Table 2: Regression Results

VARIABLES	OLS	IV-Annual Avg Wage	IV Δ_3 Population	IV Δ_3 Employment
Log Greenspace (NDVI)	0.000781 (0.000689)	-0.0767*** (0.0156)	7939*** (2479)	8387*** (2046)
Observations	9,440	9,440	6,608	6,608
Number of Counties	950	950	950	950
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
<i>First Stage:</i>		NDVI	NDVI	NDVI
Historic Land Cover		0.090*** (0.0065)	0.078*** (0.0068)	0.078*** (0.0068)
F-Stat		183.02	129.95	129.95

Standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

In the annual wage specification, the coefficient on log greenspace is negative and statistically significant at the 1% level. This indicates that greener counties have lower wage levels after accounting for endogeneity using historical land cover as an instrument. Economically, this magnitude implies that a 10% increase in NDVI lowers wages by approximately 0.77%, consistent with compensating differentials. Workers are willing to accept lower wages in exchange for higher-quality environmental amenities, as hypothesized in the Roback-style spatial equilibrium framework.

The coefficient on log greenspace is positive and statistically significant in the population change regression, consistent with greener counties attracting more residents over time. Substantively, a 1% increase in NDVI corresponds to roughly 80 additional residents over a three-year interval. This magnitude is consistent with mobility responses because mi-

gration involves frictions such as moving costs, imperfect information, and household-level constraints.

Examining the impact on employment change, log greenspace also has a positive and significant impact. This implies that a 1% increase in NDVI is associated with an increase of approximately 84 additional jobs over a three-year period. The employment response aligns with the population result: as workers relocate toward greener areas, local labor supply rises, prompting firms to expand or enter the market.

The first-stage F-statistics well exceed the conventional threshold, supporting the strength of the instrument (historic land cover) used to isolate exogenous variation in greenspace. Present NDVI is highly correlated with past land cover patterns of greenery, when controlling for present and historical industrial composition. The detailed first stage regressions are provided in the *appendix 2*.

6.2 Robustness Checks

To assess whether the estimated effects of greenspace reflect genuine underlying relationships rather than spurious correlations or mechanical patterns in the data, I implement a simple placebo test that randomizes the distribution of NDVI values across counties. Specifically, I generate a randomized version of the greenspace measure by independently permuting the NDVI values while preserving the full sample size and empirical distribution. I then re-estimate the three core IV specifications: annual wages, three-year population change, and three-year employment change. The summary results are presented in *table 3*, while detailed results are provided in *appendix 4*.

Across all three placebo regressions, the coefficient on the randomized NDVI variable becomes statistically insignificant. In other words, once the spatial structure of greenspace allocation is dismantled, none of the relationships documented in the primary analysis persist. This indicates that the wage penalty associated with greenspace and the positive effects on

Table 3: Robustness Check: Randomized NDVI Regression Results

VARIABLES	IV-Annual Avg Wage	IV Δ_3 Population	IV Δ_3 Employment
Log Randomized Greenspace (NDVI)	1.063 (1.467)	117545 (295186)	28727 (74778)
Observations	9,440	6,608	6,608
Number of Counties	950	950	950
State FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
<i>First Stage:</i>	r NDVI	r NDVI	r NDVI
Historic Land Cover	-0.0059 (0.0081)	-0.0039 (0.0099)	-0.0039 (0.0099)
F-Stat	0.53	0.16	0.16

*Standard errors in parentheses.**** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

population and employment growth are not mechanical artifacts of scale, time trends, or noise in the data. Rather, the estimated effects depend critically on the true spatial distribution of greenspace, which is consistent with the spatial-equilibrium mechanism discussed above.

Importantly, the placebo results reinforce the interpretation that greenspace is a meaningful location-specific attribute driving sorting behavior and labor-market adjustments, not a proxy for unrelated county characteristics. If the main findings were driven solely by omitted variables or random patterns, the randomized NDVI variable would have reproduced similar coefficients. Instead, the absence of any significant effect in the placebo models strengthens the internal validity of the identification strategy and provides additional confidence that the main results capture genuine causal relationships between greenspace, wages, and demographic adjustment.

7 Discussion and Policy Implications

The combined wage, population, and employment results provide a coherent picture of how urban greenspace shapes local economic dynamics through the spatial-equilibrium mecha-

nism. The results suggest that workers are willing to accept lower earnings in exchange for the higher QOL provided by greenspace amenities. This amenity-driven migration reflects a labor supply response, whereby individuals relocate to greener areas despite lower wages, consistent with spatial equilibrium models. As a result, population inflows contribute to higher employment levels in counties with greater greenspace provision. The results contribute to a growing recognition that environmental and macroeconomic objectives can be complementary: environmental amenities like greenspace not only yield health and climate benefits (e.g. heat mitigation, flood management), but also economic benefits in terms of attracting labor and supporting growth. The following is a discussion of the policy implications of these findings.

First, the findings indicate that greenspace is utility enhancing, which leads to in-migration and increases in labor supply, which in turn promotes local economic growth. In the wage equation (*equation 2*), greener counties exhibit significantly lower average wages, consistent with a compensating differential in which workers trade income for environmental amenities.

In the population and employment models (*equation 3 & 4*), counties with higher NDVI experience substantially larger three-year increases in both residents and jobs. These adjustments are consistent with the spatial-equilibrium view that wage differences transmit amenity values to workers, while population flows reveal the gradual reallocation of labor toward more desirable locations. From a broader macroeconomic prospective, urban greenspace acts as a pull factor that increases local labor supply by attracting in-migration. This inflow of workers can support higher output, employment, and tax revenues, even if wages are modest. The implication is that amenities, traditionally viewed as consumption-side benefits, may also serve as a lever for competitiveness and regional growth. For struggling areas, improving QOL amenities offers an alternative to traditional labor market interventions like wage subsidies or tax breaks. For policymakers, this implies that investments in parks, tree canopy, and urban greening projects may serve as cost-effective tools for workforce attraction and

retention, particularly in regions facing demographic stagnation or out-migration.

Second, the results help explain persistent regional variation in development. Counties with higher greenspace not only have lower wages but also display faster demographic expansion. This pattern aligns with long-standing empirical observations that amenities contribute to uneven regional development, particularly between high-amenity and low-amenity areas (Chi and Marcouiller, 2013). By demonstrating that NDVI predicts both wage levels and labor-market adjustments, this study provides evidence that environmental amenities are a structural driver of regional divergence. Regions endowed with or investing in greenspace can attract workers even at lower wage levels, while less green regions must compensate workers with higher wages to offset lower amenities. Over time, this dynamic can produce cumulative differences in population growth, employment opportunities, and local tax bases. Thus, amenity-driven economic sorting, often overlooked in traditional productivity-based regional analyses, may be a fundamental mechanism behind emerging patterns of regional inequality in the U.S.

Third, the findings support a framework for policies promoting green urban growth. Unlike static natural amenities such as coastline or mountains, greenspace is highly malleable through public policy. Urban tree planting initiatives, park creation, greening of vacant lots, and zoning reforms that require or incentivize green coverage can significantly change a county’s NDVI within a short time horizon. The empirical results imply that such interventions can reshape local labor-market conditions by attracting residents and fostering employment expansion. Importantly, this “green growth” does not rely on offering higher wages or business subsidies; rather, it enhances the underlying attractiveness of place. This suggests a powerful complement to traditional economic-development tools: enhancing environmental quality not only yields health and ecological benefits but also contributes to long-run economic vitality by expanding the workforce and stimulating job creation.

Taken together, the results indicate that greenspace is more than an aesthetic or environmental feature, it is an economic asset, much like water access and other environmental

amenities increasingly recognized as macro-critical in development policy (Suphachalasai et al., 2025). These factors can affect labor-market equilibrium, migration patterns, and contributes to uneven regional development. These findings highlight the importance of integrating greenspace into economic-development planning and suggest that urban greening efforts may yield substantial returns beyond environmental improvement. Although the analysis is U.S. based, the fundamental insights that QOL amenities influence where people choose to live and work likely holds in other contexts. City planners worldwide, especially in rapidly urbanizing economies, might draw on these findings when balancing development and livability, ensuring sustainable growth.

8 Conclusion

This paper provides new evidence that urban greenspace plays a meaningful role in shaping labor-market outcomes, demographic patterns, and local economic development. Using a panel of 950 U.S. urban counties from 2011–2019 and an IV design leveraging historical land cover, I show that workers accept lower wages in greener areas, consistent with compensating differentials predicted by spatial-equilibrium theory. At the same time, greener counties experience significantly higher three-year population and employment growth, suggesting that greenspace attracts households and stimulates labor-market expansion. Taken together, these results demonstrate that greenspace is not merely an environmental amenity but also a determinant of economic behavior and urban adjustment dynamics.

These findings also highlight the role of greenspace in producing persistent regional differences in development. High-NDVI counties exhibit strong in-migration and employment expansion even at lower wage levels, while low-NDVI counties maintain higher wages but struggle with slower growth. These patterns underscore the importance of amenities in explaining why some regions continue to expand while others stagnate despite traditional economic-development efforts. Because NDVI is malleable through public investment and

land-use planning, these results imply that local governments may influence long-run growth trajectories by enhancing greenspace coverage rather than relying solely on subsidies or business-attraction programs.

Overall, the evidence suggests that urban greenspace is an underappreciated component of economic competitiveness. Investments in parks, tree canopy, and environmental quality not only improve well-being but also influence labor supply, population flows, and employment growth. In a period when many U.S. regions face demographic decline, productivity challenges, and widening spatial disparities, understanding the economic value of greenspace offers policymakers a new tool for sustainable urban development.

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Appendix

Appendix 1: IV testing regression

Table 1: NDVI regressed with Historical Landcover

VARIABLES	ln_NDVI	ln_wage
hist_landcover	0.0898*** (0.00876)	0.000768 (0.000723)
miles to nearest MSA	0.000550 (0.000729)	-0.00147*** (0.000260)
present_landcover	7.12e-08*** (9.94e-09)	-8.27e-09** (3.74e-09)
HPI	0.000242*** (5.50e-05)	7.87e-05*** (6.24e-06)
% working in-county	-0.00131** (0.000592)	0.00121*** (0.000141)
% High School Degree	-0.00326 (0.00238)	0.00180*** (0.000316)
% Bachelors Degree	-0.00514 (0.00342)	0.00423*** (0.000446)
% Graduate Degree	0.00154 (0.00406)	0.00265*** (0.000555)
% Employed	0.00133 (0.00193)	5.81e-05 (0.000264)
Median HH income (2010 USD)	1.32e-06 (1.13e-06)	1.56e-06*** (1.68e-07)
Total Population	-7.74e-08* (4.56e-08)	-5.02e-08*** (1.27e-08)
% Black	0.00155* (0.000840)	0.00233*** (0.000281)
% American Indian/Alaskan	0.00773* (0.00436)	0.00167 (0.00105)
% Asian	-0.00591* (0.00348)	0.00550*** (0.000861)
% Hawaii/Pacific	0.0850** (0.0433)	0.00583 (0.00402)
% Other race	-0.00381 (0.00269)	-5.35e-05 (0.000271)
% Multiracial	0.0215*** (0.00662)	0.000558 (0.000692)
Annual avg precipitation	0.00417 (0.200)	0.0206 (0.0138)
NAICS - Mine, Oil, Gas	0.000813 (0.00384)	0.0162*** (0.000487)
NAICS - Utilities	0.00150 (0.00717)	0.0121*** (0.00129)

Continued on next page

Table 4 – continued from previous page

VARIABLES	ln_NDVI	ln_wage
NAICS - Construction	0.00799** (0.00323)	0.00907*** (0.000406)
NAICS - Manufacturing 1	-0.000303 (0.00206)	0.00300*** (0.000368)
NAICS - Manufacturing 2	0.00270 (0.00236)	0.00626*** (0.000399)
NAICS - Manufacturing 3	0.000394 (0.00151)	0.00612*** (0.000300)
NAICS - Wholesale	-0.00198 (0.00464)	0.00390*** (0.000630)
NAICS - Retail 1	0.00413 (0.00563)	-0.00306*** (0.000728)
NAICS - Retail 2	0.00463 (0.00378)	-0.00293*** (0.000548)
NAICS - Transport 1	-0.00169 (0.00389)	0.00431*** (0.000723)
NAICS - Transport 2	-0.00522 (0.00436)	-0.00106** (0.000506)
NAICS - Information	-0.00239 (0.00884)	0.0148*** (0.00117)
NAICS - Finance, Insurance	0.00244 (0.00329)	0.0111*** (0.000607)
NAICS - Real Estate	-0.00571 (0.0132)	-0.000235 (0.00153)
NAICS - Prof Science Tech	-0.00169 (0.00387)	0.0108*** (0.000610)
NAICS - Management	0.00940* (0.00512)	0.00824*** (0.000631)
NAICS - Admin, Waste	-0.00696** (0.00352)	0.000119 (0.000415)
NAICS - Education	0.00371 (0.00307)	0.00111* (0.000604)
NAICS - Healthcare	0.00385** (0.00184)	6.77e-05 (0.000340)
NAICS - Arts, Entertainment	0.00613* (0.00366)	-0.00122* (0.000736)
NAICS - Accommodation, Food	-0.00560* (0.00295)	-0.00673*** (0.000520)
NAICS - Other Services	-0.00933* (0.00489)	-0.000795 (0.000544)
20 hist_ind_value_70	5.26e-06*** (1.88e-06)	1.49e-06** (7.13e-07)
40 hist_ind_value_70	1.06e-05*** (3.73e-06)	1.99e-06 (1.43e-06)
50 hist_ind_value_70	2.14e-05 (1.81e-05)	-2.22e-05*** (7.17e-06)
70 hist_ind_value_70	7.15e-06 (1.03e-05)	7.12e-07 (4.02e-06)

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Table 4 – continued from previous page

VARIABLES	ln_NDVI	ln_wage
90 hist_ind_value_70	-6.21e-06*** (2.38e-06)	-1.56e-06* (9.13e-07)
100 hist_ind_value_70	-4.86e-05** (2.43e-05)	1.15e-05 (9.71e-06)
200 hist_ind_value_70	8.80e-06 (6.18e-06)	-2.47e-06 (2.41e-06)
300 hist_ind_value_70	1.91e-07 (4.85e-06)	1.52e-05*** (1.82e-06)
400 hist_ind_value_70	3.07e-06* (1.76e-06)	5.99e-07 (6.88e-07)
500 hist_ind_value_70	-1.56e-05*** (4.54e-06)	3.42e-06** (1.73e-06)
610 hist_ind_value_70	1.49e-05*** (4.09e-06)	-3.86e-06** (1.60e-06)
620 hist_ind_value_70	-1.78e-06 (3.94e-06)	-4.59e-06*** (1.56e-06)
700 hist_ind_value_70	-5.12e-06 (3.75e-06)	-4.19e-07 (1.48e-06)
910 hist_ind_value_70	-6.57e-06** (3.02e-06)	2.86e-06** (1.18e-06)
920 hist_ind_value_70	-4.51e-06** (2.10e-06)	-1.62e-06** (8.12e-07)
Constant	-1.631*** (0.207)	9.978*** (0.0387)
present_NDVI		-0.0146*** (0.00331)
Observations	9,440	
Number of geoid	950	

Standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Appendix 2: First-Stage Primary Regressions

VARIABLES	(1) Wage	(2) Population/ Employment
ln_landcover_mean	0.0815684*** (0.0060542)	0.0778572*** (0.0068299)
high_temp_days	0.0029832*** (0.0003713)	-0.0013608*** (0.0004107)
near_dist	0.000325 (0.0005335)	0.0000792 (0.000576)
present_landcover	6.98e-08*** (8.83e-09)	8.31e-08*** (7.51e-09)
rent	-4.00e-06 (0.0000328)	-0.0000339 (0.0000341)
work_incounty	-0.0015355*** (0.0004102)	-0.001295*** (0.0004478)
aland	-3.98e-11*** (2.58e-12)	-4.45e-11*** (2.73e-12)
edu_perc_hs	-0.0019692 (0.0017599)	0.0004453 (0.002019)
edu_perc_bach	-0.0032808 (0.0020911)	-0.0031067 (0.0023569)
edu_perc_grad	0.0057084* (0.0030864)	0.0078469** (0.0033005)
pop_tot_pop	-1.01e-07** (3.55e-08)	
race_perc_black	0.000948* (0.0005383)	0.0013154** (0.000611)
race_perc_amindnat	0.010313*** (0.0033067)	0.0106391*** (0.0035058)
race_perc_asian	-0.0055865* (0.0031858)	-0.0019743 (0.0026665)

VARIABLES	(1) Wage	(2) Popula- tion/Employment
race_perc_hawaii	0.046304 (0.0397038)	0.0516038 (0.0328687)
race_perc_others	-0.0039113 (0.0023846)	-0.0072515*** (0.0027328)
race_perc_multirace	0.0183551*** (0.0052104)	0.0125756** (0.0057167)
precipitation	1.024296*** (0.2361019)	0.0581111 (0.2317751)
naics2_21	-0.0002753 (0.0031795)	-0.0029168 (0.0037369)
naics2_22	0.0017748 (0.0042955)	0.0001227 (0.0053111)
naics2_23	0.0078193*** (0.002269)	0.0029472 (0.0024879)
naics2_31	0.0004169 (0.0013355)	0.0004611 (0.0015076)
naics2_32	0.0035249** (0.0016458)	0.002029 (0.0019699)
naics2_33	0.0016852 (0.0012407)	0.0020864* (0.0011187)
naics2_42	-0.0000719 (0.0032704)	-0.000996 (0.0035601)
naics2_44	0.0106408*** (0.0039463)	0.0080503* (0.0042331)
naics2_45	0.0031501 (0.0023676)	0.0034556 (0.0028174)
naics2_48	0.0009212 (0.0025248)	0.0002442 (0.0030247)
naics2_49	-0.0020323 (0.0025661)	-0.0026222 (0.0027732)

VARIABLES	(1) Wage	(2) Popula- tion/Employment
naics2_51	-0.0030885 (0.0063589)	-0.0090761 (0.0069016)
naics2_52	0.0036663* (0.0021621)	0.0030238 (0.0025092)
naics2_53	0.0018184 (0.0086061)	0.0098728 (0.0096062)
naics2_54	0.0028907 (0.0024584)	0.0031403 (0.0028448)
naics2_55	0.0093361*** (0.0032515)	0.0064006 (0.0041707)
naics2_56	-0.0036135 (0.0024134)	-0.0023384 (0.0027174)
naics2_61	0.0046124** (0.0023196)	0.0028872 (0.0026725)
naics2_62	0.0023206* (0.0013321)	0.0005053 (0.0015079)
naics2_71	0.0061431** (0.0024228)	0.0059822** (0.0026107)
naics2_72	-0.0018572 (0.0014541)	-0.0033927** (0.0016586)
naics2_81	-0.0137834** (0.0063205)	-0.0005109 (0.0042495)
hist_ind_value_7020	5.17e-06*** (1.42e-06)	4.26e-06** (1.51e-06)
hist_ind_value_7040	0.0000135*** (3.16e-06)	8.17e-06*** (3.00e-06)
hist_ind_value_7050	0.0000163 (0.0000145)	0.0000285* (0.0000158)
hist_ind_value_7060	0 (omitted)	0 (omitted)
hist_ind_value_7070	0.0000109	-1.78e-06

VARIABLES	(1) Wage	(2) Popula- tion/Employment
	(8.09e-06)	(8.53e-06)
hist_ind_value_7080	0 (omitted)	0 (omitted)
hist_ind_value_7090	-6.88e-06***	-6.13e-06***
	(1.97e-06)	(2.06e-06)
hist_ind_value_70100	-0.0000532**	-0.0000328*
	(0.0000192)	(0.0000198)
hist_ind_value_70200	8.23e-06*	4.74e-06
	(4.90e-06)	(5.66e-06)
hist_ind_value_70300	1.56e-06	-1.22e-06
	(3.89e-06)	(4.57e-06)
hist_ind_value_70400	3.27e-06**	3.47e-06**
	(1.43e-06)	(1.63e-06)
hist_ind_value_70500	-0.0000112***	-7.13e-06*
	(4.09e-06)	(4.14e-06)
hist_ind_value_70610	0.0000122***	0.0000117***
	(3.43e-06)	(3.94e-06)
hist_ind_value_70620	-9.41e-07	-3.47e-07
	(2.89e-06)	(3.10e-06)
hist_ind_value_70700	-3.04e-06	-2.18e-06
	(3.22e-06)	(3.55e-06)
hist_ind_value_70800	0 (omitted)	0 (omitted)
hist_ind_value_70900	0 (omitted)	0 (omitted)
hist_ind_value_70910	-5.71e-06**	-3.62e-06*
	(2.21e-06)	(2.41e-06)
hist_ind_value_70920	-4.51e-06***	-4.33e-06***
	(1.50e-06)	(1.55e-06)
hist_ind_value_70930	0 (omitted)	0 (omitted)
Constant	-1.829727***	-1.969703***
	(0.1584742)	(0.1794386)

VARIABLES	(1) Wage	(2) Population/ Employment
Observations	9,440	6,608
State FE	Yes	Yes
Year FE	Yes	Yes

Notes: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix 3: Second-Stage Primary Regressions

VARIABLES	(1) Wage	(2) Pop	(3) Emp
ln_yndvi	-0.076702*** (0.0156273)	7,938.857*** (2,479.500)	8,387.437*** (2,046.868)
high_temp_days	0.0002417*** (0.0000831)	21.12622* (11.79154)	32.87932*** (9.742163)
miles to nearest MSA	-0.0004925*** (0.0001114)	60.46361*** (12.19620)	50.81066*** (10.05154)
present_landcover	-3.59e-09* (1.84e-09)	-0.0015998*** (0.0004619)	-0.0015267*** (0.0003665)
% of people working in-county	0.0011539*** (0.0000886)	-82.93678*** (10.29946)	-26.80409*** (9.727581)
County land area (sq km)	-1.51e-12* (8.47e-13)	8.31e-07*** (1.48e-07)	9.29e-07*** (1.44e-07)
% of population with Highschool Degree	0.0012831*** (0.0003927)	-152.9311*** (58.35699)	-66.75191 (45.70157)
% of population with Bachelors Degree	0.005017*** (0.0005163)	853.8433*** (65.43910)	479.4633*** (56.32444)
% of population with Graduate Degree	0.006209*** (0.0007611)	-1,212.461*** (96.57771)	-808.7068*** (84.31330)
Total Population	-3.30e-08*** (6.01e-09)		
% of people Black	0.0025447*** (0.0001222)	-165.5978*** (18.80722)	-134.4626*** (15.67700)
% of people American Indian/Alaskan	0.0028497*** (0.0006146)	-224.8996** (108.7427)	-251.583*** (88.10834)
% of people Asian	0.0037843*** (0.0006765)	1,297.363*** (119.4515)	1,001.920*** (98.19057)
% of people Hawaii/Pacific	0.0390387*** (0.0086973)	3,030.703*** (919.2295)	2,926.924*** (920.6672)

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VARIABLES	(1) Wage	(2) Pop	(3) Emp
% of people other race	0.0021548*** (0.0004404)	169.541** (85.17785)	10.64546 (67.29606)
% of people multiracial	0.008235*** (0.0014771)	168.0051 (128.6047)	-321.7333*** (115.6074)
Annual average precipitation	0.186533*** (0.0456549)	12,362.73** (5,944.284)	7,705.646 (4,979.286)
NAICS - Mine,Oil,Gas	0.0146633*** (0.0008655)	-345.7773*** (92.40831)	27.71008 (93.41069)
NAICS - Utilities	0.0279499*** (0.0009947)	146.1649* (81.04145)	93.58461 (72.22121)
NAICS - Construction	0.0058666*** (0.0007047)	-119.4455* (64.07526)	-2.915295 (51.22738)
NAICS - Manufacturing 1	0.0021516*** (0.0003040)	4.724069 (30.30098)	41.93714 (26.26730)
NAICS - Manufacturing 2	0.0105162*** (0.0004179)	59.18835* (31.00349)	60.11982** (26.86881)
NAICS - Manufacturing 3	0.0093084*** (0.0003212)	80.81112*** (22.97335)	115.266*** (20.31910)
NAICS - Wholesale	0.0058006*** (0.0007641)	-136.1751* (82.69778)	233.0429*** (74.01426)
NAICS - Retail 1	-0.0055381*** (0.0009981)	235.4442*** (87.89279)	-79.04325 (77.12734)
NAICS - Retail 2	-0.0024411*** (0.0007455)	-65.76329 (55.87606)	-68.64487 (45.46510)
NAICS - Transport 1	0.0085738*** (0.0005044)	272.4411*** (69.40407)	257.6663*** (55.28777)
NAICS - Transport 2	0.0032714*** (0.0005776)	80.7196 (53.17089)	234.8158*** (48.72890)
NAICS - Information	0.0136156***	1,661.334***	1,990.303***

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VARIABLES	(1) Wage	(2) Pop	(3) Emp
	(0.0017965)	(248.0162)	(244.3902)
NAICS - Finance, Insurance	0.0102738***	-66.10105	-64.40246
	(0.0006928)	(59.30346)	(54.06990)
NAICS - Real Estate	0.0036429	509.0431*	65.92996
	(0.0023518)	(265.3844)	(194.8716)
NAICS - Prof Science Tech	0.0225147***	-75.88412	263.0983***
	(0.0006023)	(77.42865)	(65.45664)
NAICS - Management	0.0252499***	-117.8982	-51.9307
	(0.0011782)	(107.7069)	(90.49143)
NAICS - Admin, Waste	0.0017817***	261.1505***	302.4747***
	(0.0006257)	(57.16289)	(55.43956)
NAICS - Education	0.000396	202.9462***	148.9646***
	(0.0003858)	(51.50378)	(44.68830)
NAICS - Healthcare	0.0026138***	0.5633516	-42.1329
	(0.0003617)	(34.57277)	(29.14365)
NAICS - Arts, Entertainment	-0.0003888	114.1653**	131.6044**
	(0.0006973)	(56.50600)	(60.87033)
NAICS - Accommodation,Food	-0.003576***	297.9094***	210.1317***
	(0.0004174)	(39.57756)	(34.46980)
NAICS - Other Services	-0.0122183***	-411.9346***	-432.9641***
	(0.0010037)	(94.46012)	(82.47425)
20 value_70	8.68e-07***	0.1923485**	0.1471738**
	(2.66e-07)	(0.0745046)	(0.0617455)
40 value_70	6.72e-07	0.0422285	0.3383326***
	(6.37e-07)	(0.1413406)	(0.1175716)
50 value_70	-4.97e-06**	0.7394243	-0.8318294*
	(2.34e-06)	(0.4925024)	(0.4781892)
70 value_70	-2.72e-06**	0.4595767	0.5086958*
	(1.35e-06)	(0.3540160)	(0.3083742)

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VARIABLES	(1) Wage	(2) Pop	(3) Emp
90 value_70	-2.84e-07 (3.44e-07)	-0.0698059 (0.1006525)	-0.1517777** (0.0772277)
100 value_70	1.37e-05*** (3.25e-06)	2.545332** (0.9932736)	1.437961* (0.7674118)
200 value_70	3.59e-06*** (7.78e-07)	1.401218*** (0.3076720)	0.0151707 (0.3968654)
300 value_70	5.11e-06*** (6.49e-07)	3.046149*** (0.2735360)	1.340023*** (0.3014683)
400 value_70	2.63e-07 (2.43e-07)	-0.3798149*** (0.0809316)	-0.1783326** (0.0688863)
500 value_70	-9.93e-08 (7.80e-07)	-0.6525973*** (0.1765965)	-0.3937755*** (0.1441432)
610 value_70	-7.03e-07 (5.65e-07)	0.048133 (0.1874015)	0.1413104 (0.1759780)
620 value_70	-4.68e-06*** (5.54e-07)	-0.5859802*** (0.1793019)	-0.1809383 (0.1684020)
700 value_70	-1.19e-06** (5.26e-07)	0.0185704 (0.1691361)	0.7103702*** (0.1761861)
910 value_70	2.03e-06*** (4.06e-07)	-0.2312665** (0.1072117)	-0.4435372*** (0.0856180)
920 value_70	-1.02e-06*** (2.77e-07)	0.043128 (0.0794500)	0.0452908 (0.0634904)
Constant	9.692074*** (0.0408754)	21,056.30*** (5,776.232)	15,199.29*** (4,713.468)
Observations	9,440	6,608	6,608

Notes: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix 4: Randomized NDVI Robustness Check

First-Stage Regressions

VARIABLES	(1) Wage	(2) Population/Employment
ln_landcover_mean	-0.0058831 (0.0081023)	-0.0039161 (0.0099060)
high_temp_days	0.000767* (0.0004487)	0.0001699 (0.0005919)
near_dist	-0.0001333 (0.0006819)	-0.0001568 (0.0008054)
present_landcover	7.92e-09 (9.11e-09)	1.32e-08 (1.06e-08)
rent	-0.0000594 (0.0000492)	-0.0000490 (0.0000575)
work_incounty	0.0002839 (0.0005169)	0.0004979 (0.0006134)
aland	7.54e-13 (2.65e-12)	-3.21e-12 (3.38e-12)
edu_perc_hs	-0.0021633 (0.0022679)	0.0006177 (0.0027955)
edu_perc_bach	0.0050499* (0.0028465)	0.0053913 (0.0034595)
edu_perc_grad	-0.0080862* (0.0041270)	-0.0047921 (0.0049315)
pop_tot_pop	4.57e-08 (4.18e-08)	7.19e-08 (4.96e-08)
race_perc_black	0.0001758 (0.0007573)	0.0007224 (0.0008985)
race_perc_amindnat	-0.0013411 (0.0055080)	0.0032660 (0.0046627)

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VARIABLES	(1) Wage	(2) Popula- tion/Employment
race_perc_asian	0.0022297 (0.0034879)	0.0003441 (0.0042391)
race_perc_hawaii	0.0513698 (0.0478560)	0.0485410 (0.0569519)
race_perc_others	0.0018561 (0.0030075)	0.0040044 (0.0034831)
race_perc_multirace	-0.0044544 (0.0070533)	-0.0103449 (0.0083628)
precipitation	0.2259089 (0.2625121)	0.0951743 (0.3107639)
naics2_21	0.0009505 (0.0042658)	-0.0000975 (0.0055745)
naics2_22	-0.0069530 (0.0070430)	-0.0102408 (0.0093816)
naics2_23	0.0049295 (0.0030070)	0.0029187 (0.0036198)
naics2_31	-0.0021133 (0.0021670)	-0.0016580 (0.0027665)
naics2_32	0.0002134 (0.0020922)	-0.0007034 (0.0024233)
naics2_33	-0.0040223** (0.0016793)	-0.0030941 (0.0020874)
naics2_42	-0.0043695 (0.0045748)	-0.0028225 (0.0056244)
naics2_44	0.0109597* (0.0056244)	0.0107866 (0.0066921)
naics2_45	-0.0069182** (0.0034323)	-0.0052907 (0.0041658)

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VARIABLES	(1) Wage	(2) Population/ Employment
naics2_48	0.0035024 (0.0032574)	0.0095103** (0.0038280)
naics2_49	0.0003948 (0.0042676)	-0.0014757 (0.0046501)
naics2_51	-0.0108080 (0.0090167)	-0.0147974 (0.0113725)
naics2_52	-0.0095113*** (0.0036057)	-0.0105849** (0.0044404)
naics2_53	0.0002005 (0.0140802)	0.0002636 (0.0147979)
naics2_54	-0.0015951 (0.0035560)	-0.0012575 (0.0041670)
naics2_55	0.0010181 (0.0055030)	0.0021995 (0.0068823)
naics2_56	-0.0019251 (0.0036357)	-0.0012432 (0.0042069)
naics2_61	-0.0011664 (0.0028929)	-0.0018306 (0.0034617)
naics2_62	-0.0003020 (0.0017434)	-0.0012024 (0.0020448)
naics2_71	-0.0029322 (0.0047152)	-0.0060295 (0.0064812)
naics2_72	0.0012748 (0.0019668)	0.0033997 (0.0021448)
naics2_81	-0.0068893 (0.0047102)	0.0001773 (0.0058652)
hist_ind_value_7020	5.35e-08 (1.70e-06)	-9.49e-07 (2.04e-06)

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VARIABLES	(1) Wage	(2) Popula- tion/Employment
hist_ind_value_7040	-1.13e-06 (3.63e-06)	-5.64e-06 (4.38e-06)
hist_ind_value_7050	0.0000263 (0.0000172)	0.0000289 (0.0000204)
hist_ind_value_7060	0 (omitted)	0 (omitted)
hist_ind_value_7070	-0.0000241** (0.0000101)	-0.0000240* (0.0000123)
hist_ind_value_7080	0 (omitted)	0 (omitted)
hist_ind_value_7090	5.67e-08 (2.24e-06)	1.45e-06 (2.78e-06)
hist_ind_value_70100	0.0000349 (0.0000262)	0.0000489* (0.0000291)
hist_ind_value_70200	-6.70e-06 (5.95e-06)	-9.54e-06 (7.21e-06)
hist_ind_value_70300	2.81e-06 (4.86e-06)	4.87e-06 (5.84e-06)
hist_ind_value_70400	-5.55e-07 (1.69e-06)	-4.42e-07 (2.05e-06)
hist_ind_value_70500	-1.41e-06 (4.51e-06)	-3.30e-06 (5.66e-06)
hist_ind_value_70610	3.85e-06 (3.96e-06)	1.43e-06 (4.86e-06)
hist_ind_value_70620	-3.82e-07 (3.75e-06)	-1.68e-07 (4.43e-06)
hist_ind_value_70700	-4.37e-06 (3.90e-06)	-3.55e-06 (4.60e-06)
hist_ind_value_70800	0 (omitted)	0 (omitted)
hist_ind_value_70900	0 (omitted)	0 (omitted)

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VARIABLES	(1) Wage	(2) Popula- tion/Employment
hist_ind_value_70910	2.84e-06 (2.74e-06)	4.85e-06 (3.27e-06)
hist_ind_value_70920	2.52e-07 (1.96e-06)	8.49e-07 (2.42e-06)
hist_ind_value_70930	0 (omitted)	0 (omitted)
Observations	9,440	6,608
State FE	Yes	Yes
Year FE	Yes	Yes

Notes: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix 4: Second-Stage Regressions with Randomized NDVI

VARIABLES	(1) Wage	(2) Population	(3) Employment
ln_yndvi_rand	1.063467 (1.467913)	117545.2 (295186.6)	28727.1 (74778.93)
high_temp_days	-0.0008028 (0.0011983)	-16.89981 (83.75258)	11.43764 (21.59312)
near_dist	-0.0003757 (0.0007586)	15.79733 (105.3854)	10.74143 (26.92816)
present_landcover	-1.74e-08 (1.35e-08)	-0.0007651 (0.0038107)	0.0000174 (0.0009658)
rent	0.0002054** (0.0000987)	3.535098 (15.69455)	-0.0139065 (4.008463)
work_incounty	0.0009698 (0.0006767)	-71.13606 (158.6924)	5.252745 (40.92456)
aland	7.44e-13 (3.12e-12)	-5.75e-08 (1.03e-06)	5.60e-10 (2.63e-07)
edu_perc_hs	0.0037347 (0.0040333)	-267.9447 (368.7653)	-113.3711 (95.18902)
edu_perc_bach	-0.0001018 (0.0082533)	-480.9974 (1676.269)	-181.6538 (427.4357)
edu_perc_grad	0.0143706 (0.0128076)	83.98784 (1542.721)	-129.0088 (392.8249)
pop_tot_pop	-7.39e-08 (7.58e-08)	0.0483965** (0.0213277)	0.0382908*** (0.0055078)
race_perc_black	0.002285*** (0.0008690)	-124.9497 (244.0535)	-62.4614 (61.91073)
race_perc_amindnat	0.0034849 (0.0061499)	-287.5529 (1134.151)	-88.08659 (286.6142)
race_perc_asian	0.0018415 (0.0050086)	525.3455 (503.8490)	467.2922*** (150.7589)

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VARIABLES	(1) Wage	(2) Population	(3) Employment
race_perc_hawaii	-0.0191431 (0.0907077)	-5812.703 (15792.16)	-552.7825 (4030.799)
race_perc_others	0.0004809 (0.0041252)	-481.6821 (1231.710)	-252.4909 (313.3963)
race_perc_multirace	0.0115642 (0.0100083)	1497.935 (3222.528)	90.9297 (819.6269)
precipitation	-0.1322793 (0.4159672)	-9819.516 (43980.14)	-2673.358 (11557.11)
naics2_21	0.0136735*** (0.0046105)	-30.79637 (654.4715)	237.9372 (179.3328)
naics2_22	0.0352081*** (0.0127662)	1317.16 (3236.364)	364.8575 (814.3111)
naics2_23	0.0000245 (0.0076260)	-429.0185 (914.6694)	-54.8659 (233.9341)
naics2_31	0.0043671 (0.0038987)	206.5988 (589.0190)	95.79472 (147.6391)
naics2_32	0.010019*** (0.0022528)	64.89479 (363.1504)	31.26645 (90.81094)
naics2_33	0.0134567** (0.0059555)	370.281 (912.3224)	157.2004 (230.5123)
naics2_42	0.010453 (0.0077551)	64.23184 (1030.714)	218.1326 (261.6909)
naics2_44	-0.0180096 (0.0170492)	-1507.763 (3274.715)	-704.1471 (828.6831)
naics2_45	0.0046745 (0.0108868)	522.1657 (1640.670)	68.74094 (415.6541)
naics2_48	0.0047785 (0.0061674)	-1132.364 (2830.426)	-218.5324 (718.6803)
naics2_49	0.0030074	130.7205	182.3512

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VARIABLES	(1) Wage	(2) Population	(3) Employment
	(0.0045823)	(675.1109)	(170.0037)
naics2_51	0.0253464	2653.649	1860.115
	(0.0190203)	(4654.292)	(1177.837)
naics2_52	0.0201076	1205.134	267.1796
	(0.0144602)	(3163.598)	(799.9737)
naics2_53	0.0032901	695.6243	239.9686
	(0.0150011)	(1757.289)	(443.9414)
naics2_54	0.0239893***	131.3865	350.0709**
	(0.0045522)	(631.5667)	(162.4349)
naics2_55	0.023451***	-140.723	69.82301
	(0.0061914)	(1013.558)	(250.3762)
naics2_56	0.0041062	-34.64079	18.04976
	(0.0048359)	(601.5295)	(155.2150)
naics2_61	0.0012826	274.9457	107.8620
	(0.0035283)	(685.9030)	(172.5639)
naics2_62	0.002757	83.47664	-47.67442
	(0.0019900)	(434.0061)	(109.8322)
naics2_71	0.0022584	764.9343	280.1292
	(0.0066444)	(1950.977)	(493.2416)
naics2_72	-0.0047893*	-310.5721	-45.13288
	(0.0027403)	(1006.186)	(255.0542)
naics2_81	-0.0038345	45.10384	-100.2402
	(0.0114003)	(686.4047)	(173.7228)
hist_ind_value_7020	4.15e-07	-0.0435402	-0.060502
	(1.81e-06)	(0.3923073)	(0.1041933)
hist_ind_value_7040	8.35e-07	-0.0758927	-0.0314723
	(4.33e-06)	(1.788424)	(0.4554073)
hist_ind_value_7050	-0.0000342	-2.26152	-1.301915
	(0.0000457)	(9.242763)	(2.342129)

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VARIABLES	(1) Wage	(2) Population	(3) Employment
hist_ind_value_7060	0 (omitted)	0 (omitted)	0 (omitted)
hist_ind_value_7070	0.0000221 (0.0000374)	3.075072 (7.230316)	1.047499 (1.833578)
hist_ind_value_7080	0 (omitted)	0 (omitted)	0 (omitted)
hist_ind_value_7090	1.84e-07 (2.39e-06)	0.0281949 (0.5720132)	-0.0199231 (0.1508211)
hist_ind_value_70100	-0.0000194 (0.0000593)	-6.710405 (14.84232)	-2.549677 (3.759787)
hist_ind_value_70200	0.0000101 (0.0000116)	1.348515 (2.939937)	-0.5311089 (0.8638526)
hist_ind_value_70300	2.01e-06 (6.66e-06)	0.5832925 (1.607278)	-0.1450347 (0.4714919)
hist_ind_value_70400	6.02e-07 (1.99e-06)	-0.1604629 (0.2809537)	-0.0372765 (0.0854173)
hist_ind_value_70500	2.25e-06 (5.21e-06)	0.4490651 (1.178418)	0.1878483 (0.3093996)
hist_ind_value_70610	-5.74e-06 (6.62e-06)	0.3502479 (0.6599811)	0.4662408** (0.2059619)
hist_ind_value_70620	-4.20e-06 (4.00e-06)	-0.6754001 (0.5268831)	-0.2545343 (0.1773014)
hist_ind_value_70700	3.69e-06 (7.46e-06)	0.3808051 (1.150357)	0.7673239** (0.3184521)
hist_ind_value_70800	0 (omitted)	0 (omitted)	0 (omitted)
hist_ind_value_70900	0 (omitted)	0 (omitted)	0 (omitted)
hist_ind_value_70910	-5.46e-07 (5.48e-06)	-0.3314253 (1.557081)	-0.2593944 (0.3941761)
hist_ind_value_70920	-9.40e-07 (2.13e-06)	-0.0068883 (0.3921345)	0.0443192 (0.1053452)
hist_ind_value_70930	0 (omitted)	0 (omitted)	0 (omitted)

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VARIABLES	(1) Wage	(2) Population	(3) Employment
year dummies	Yes	Yes	Yes
state dummies	Yes	Yes	Yes
Observations	9,440	6,608	6,608

Notes: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.